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Mitigating cognitive offloading in GenAI-supported collaboration: a metacognitive scaffolding approach to enhancing students' critical thinking

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ABSTRACT

While Generative Artificial Intelligence (GenAI) has transformative potential for collaborative learning, it may also induce cognitive offloading and superficial consensus, leading learners to accept AI-generated outputs uncritically. To address these risks, this study developed metacognitive scaffolds for triadic "Learner–Peer–GenAI" interaction to foster critical thinking. An eight-week quasi-experimental intervention was conducted with 85 sixth-grade students, comparing an experimental group receiving scaffolds (n = 42) with a control group without scaffolding (n = 43). Using a multi-method design involving pre/post tests, SOLO-based cognitive-structure analysis, Epistemic Network Analysis (ENA), and Lag Sequential Analysis (LSA), we examined changes in critical-thinking performance, cognitive structure, and interaction trajectories. Results showed that the experimental group significantly outperformed the control group in overall critical thinking and three dimensions: questioning and verification, logical reasoning, and reflection and regulation, with the largest gains in questioning and verification. High-order cognitive structures accounted for 49.7% of the experimental group's discourse, indicating a shift toward higher-level thinking. Behavioral analyses identified a dominant sequence—Questioning (QB.AI)→Evidence Retrieval (ER.AI)→Logical Analysis (LA.AI)—and a more cohesive epistemic network, whereas the control group showed surface-level exchange and passive agreement. These findings suggest that metacognitive scaffolding can regulate GenAI-supported collaboration, preserve learner agency, and promote cognitive reciprocity.

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Generative artificial intelligence (GenAI); metacognitive scaffolding; collaborative learning; critical thinking; epistemic network analysis; lag sequential analysis

1. Introduction

As Generative Artificial Intelligence (GenAI) becomes deeply embedded within educational contexts, its function in collaborative learning has transmuted from a peripheral auxiliary tool into a highly interactive "artificial cognitive agent." Nonetheless, the integration of GenAI does not intrinsically guarantee improved collaborative quality (Khosravi et al., 2025; S. Lee et al., 2025); rather, it has precipitated growing concerns regarding AI-related cognitive offloading and epistemic overreliance. The perceived fluency and authoritative tone of AI-generated outputs often trigger a "mental autopilot" state in learners (Cananau et al., 2025; Han et al., 2025), where critical socio-cognitive conflicts – essential for effective collaboration – are supplanted by superficial consensus (Song et al., 2025). This technology-induced inertia compels students toward "cognitive offloading," effectively outsourcing complex intellectual tasks to AI and eroding the deep reflective processes fundamental to collaborative learning (Taggart & Wheeler, 2025; Wu et al., 2025).

While existing literature underscores the efficacy of metacognitive scaffolding in fostering critical thinking within traditional collaborative settings, these findings are contested in GenAI-mediated environments (H. Kim et al., 2025; Molenaar et al., 2014). A critical synthesis of current research indicates that most scaffolding frameworks are contingent upon dyadic "human-to-human" interaction models designed to regulate interpersonal dissent (M. Zhou & Lam, 2019; Tedla & Chen, 2025). However, in "human-AI" or "human-human-AI" triadic contexts, learners engage with an entity defined by inherent logical and structural opacity (P. Wang et al., 2025). Traditional scaffolding may become less effective in GenAI-supported collaboration, not simply because the context has changed, but because it is not designed to regulate learners' responses to AI-generated outputs. Since such outputs are often fluent, plausible, and seemingly authoritative, learners may

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accept them too readily without sufficient questioning or verification (H.-S. Wang et al., 2021). More specifically, when scaffolding focuses mainly on interpersonal coordination or general discussion support, it may not adequately address the new epistemic challenges introduced by GenAI. As a result, learners may be more likely to accept AI outputs heuristically, overlook the need for verification, or shift evaluative judgment to the system (Lissitsa, 2025; Msambwa et al., 2025).

This study argues that GenAI should not be viewed merely as a background condition, but as a factor that metacognitive interventions need to address explicitly (Vo et al., 2024). On this basis, we redesigned the metacognitive scaffolding for GenAI-supported environments by shifting its role from general discussion support to the regulation of learners' engagement with AI-generated outputs. Specifically, the scaffolding was designed to reduce uncritical reliance on AI, promote evidence verification, and encourage logical examination of AI-generated information (Ge et al., 2025; Kaliisa et al., 2025). The scaffolding is deliberately engineered to cultivate a robust "Questioning–Evidence Retrieval–Logical Analysis" behavioral loop, specifically tailored to scrutinize AI-generated content (J. Y. Kim & Lim, 2019). The theoretical significance of this design stems from its functional specificity; the removal of the GenAI variable would render the scaffolding substantially less meaningful, as its primary mechanism targets dependence-related interaction patterns associated with AI-generated outputs, thereby affirming GenAI's indispensable role within this theoretical framework (Kong et al., 2025; Molenaar, Van Boxtel, et al., 2011).

Grounded in this rationale, the present eight-week quasi-experimental study examines how metacognitive scaffolding (specifically calibrated for GenAI environments) reconfigures the behavioral trajectories of critical thinking among primary school students. The theoretical contributions of this work are twofold: First, it interrogates the "technological neutrality" fallacy by elucidating the mechanisms underlying critical thinking impairment within GenAI-supported contexts. Second, it provides empirical evidence demonstrating how metacognitive scaffolding re-establishes cognitive equilibrium in triadic "human-human-AI" interactions through the deconstruction of algorithmic logic. This research not only introduces a novel intervention paradigm for the GenAI era but also offers a robust empirical foundation for the conceptual evolution of critical thinking theory in the age of intelligence.

2. Literature review

2.1. Critical thinking in collaborative learning

Critical thinking, conceptualized as a pivotal competency in collaborative learning for fostering in-depth knowledge construction and higher-order cognitive development (Fan et al., 2025; Ku & Hool, 2024), represents a sophisticated cognitive process involving the active questioning, systematic analysis, critical evaluation, and dynamic reconstruction of information within social interactions (Imjai et al., 2025). In specific collaborative contexts, this competency is demarcated by three interrelated core dimensions. First, evidence awareness entails the capacity to verify the authenticity, relevance, and reliability of peer contributions or GenAI-generated content against factual and logical benchmarks (Hu et al., 2025; Li et al., 2025). Second, logical reasoning emphasizes the ability to decipher causal or conditional relationships among disparate perspectives and to detect inconsistencies within the argumentative chain (García-Carmona, 2025). Finally, reflective reconstruction is manifested as learners critically scrutinize their own cognitive biases, recalibrate inquiry strategies, and synthesize diverse viewpoints to forge a coherent cognitive structure (Guo et al., 2024). These dimensions synergistically underpin the operational mechanism of critical thinking in collaborative environments.

Despite substantial empirical evidence establishing a positive correlation between collaborative learning and critical thinking (S. S. Lee & Moore, 2024), existing scholarship remains predominantly focused on outcome-oriented validation at the expense of elucidating the underlying cognitive and regulatory mechanisms (Hou et al., 2025; Lima et al., 2025). Specifically, the literature lacks a granular understanding of how critical thinking functions as a dynamic, evolving process when collaborative inquiry is increasingly mediated by advanced technologies (Leibovitch et al., 2025). As knowledge production becomes progressively decoupled from human agency, deciphering how learners calibrate their cognitive engagement and evaluative judgment presents a critical theoretical imperative (Fahrni et al., 2025). These lacunae underscore the necessity of shifting the research focus from confirming the efficacy of collaborative learning to interro-

gating how its generative pathways are reconfigured within emerging technology-mediated ecosystems (Liu & Zhong, 2025; Pozuelos et al., 2019).

2.2. Generative AI–supported collaborative learning

Recent advances in generative artificial intelligence (GenAI) have accelerated its adoption in collaborative learning environments (Galindo-Domínguez et al., 2025). Prior research has predominantly examined GenAI as an assistive technology that enhances learning efficiency by providing explanations, feedback, summaries, or language support (Sánchez Muñoz et al., 2025; Yan et al., 2025). From this perspective, GenAI is typically conceptualized as an instrumental resource that supplements human interaction and facilitates task completion (Molenaar, Chiu, et al., 2011). However, emerging scholarship suggests that GenAI differs qualitatively from earlier educational technologies. Unlike tools that merely mediate access to information, GenAI actively generates epistemic content, including arguments, interpretations, and problem solutions, which learners may appropriate, adapt, or contest during collaboration (Askell-Williams et al., 2012). This generative capacity introduces a distinctive epistemic condition in which learners increasingly engage with knowledge that is externally produced and whose underlying reasoning processes and assumptions are not fully transparent (Molenaar et al., 2010; Roll et al., 2012).

Consequently, GenAI transcends the role of a neutral auxiliary; instead, it increasingly functions as an active epistemic agent that co-constructs knowledge within collaborative environments (Ospankulova et al., 2025). Learners are tasked not only with integrating peer perspectives but also with interrogating the credibility, relevance, and applicability of AI-generated outputs. This shift fundamentally reconfigures collaborative cognition by diversifying the epistemic actors participating in meaning-making (Rehman et al., 2025). Despite burgeoning interest in GenAI-supported learning, existing scholarship remains predominantly preoccupied with learning outcomes or user perceptions, leaving the procedural intricacies of cognitive and regulatory dynamics largely underexplored (Gammoh, 2025; Sun et al., 2025). Specifically, there is a profound lack of empirical evidence regarding how learners modulate their cognition when AI-generated knowledge becomes an irreducible element of collaborative discourse (Pereles et al., 2024; Schwartz et al., 2021). This gap necessitates theoretically grounded inquiries that treat GenAI not as a peripheral enhancement, but as a transformative epistemic force that restructures the very architecture of collaborative learning.

2.3. Metacognitive scaffolding in AI-mediated learning contexts

Metacognitive scaffolding has long been established as a robust instructional framework for supporting learners' planning, monitoring, and evaluation within collaborative settings (Rehman et al., 2023). Extensive scholarship has demonstrated that metacognitive prompts and regulatory guidance effectively catalyze critical thinking and knowledge construction, contributing to sophisticated models of socially shared regulation (Ahmadi Safa & Motaghi, 2024; Yildiz Durak et al., 2025). Notably, however, prevailing frameworks rest upon a foundational epistemic assumption: that the objects of regulation are exclusively endogenous (the learner's own cognition) or intersubjective (the contributions of human peers). In these traditional contexts, the provenance and reasoning trajectories of knowledge are relatively transparent, allowing learners to exercise regulatory control through established modes of self-reflection and social negotiation (Rana et al., 2025; Stenseth et al., 2025).

The emergence of GenAI fundamentally destabilizes this traditional assumption (Matsuda et al., 2020). When AI systems synthesize explanations or solutions, learners must navigate beyond cognitive strategy regulation to actively calibrate their epistemic stance toward algorithmically-generated content. This involves critical decisions regarding the veracity of AI outputs, the necessity of cross-verification, and the strategic integration of such content into collaborative reasoning (Samala et al., 2025). Consequently, the scope of metacognitive regulation expands to encompass a triadic interaction with a non-human epistemic agent. Existing empirical evidence suggests that, in GenAI-supported environments, structured scaffolding can reorganize the cognitive relations underlying human–AI interaction (T. Wang et al., 2025; Yu et al., 2025), thereby offering further support for the present study's examination of how metacognitive scaffolding shapes triadic Learner–Peer–GenAI interaction. Despite the established importance of metacognitive scaffolding, current research has seldom interrogated how these regulatory mechanisms are reconfigured under

GenAI mediation (Suriano et al., 2025). Most inquiries remain tethered to evaluating learning outcomes, overlooking how scaffolding reorients learners' regulatory focus in response to GenAI (Anthonysamy et al., 2024). This gap necessitates a reconceptualization of metacognitive scaffolding as a dynamic process capable of addressing the epistemic transformations inherent in generative artificial intelligence.

2.4. Research question

To address the above theoretical and empirical gap, this study examines the role of metacognitive scaffolding in supporting primary school students' critical engagement in GenAI-supported collaborative learning. It focuses on a more clearly bounded question: whether, within a specific primary-school Learner–Peer–GenAI collaborative context, metacognitive scaffolding helps redirect students' regulatory activity from general task completion and interpersonal coordination toward questioning, evidence verification, logical analysis, and reflective adjustment in response to GenAI outputs.

To clarify the scope and boundaries of the research objectives, the study investigates this issue across three analytically related levels: learning outcomes, cognitive engagement, and interaction processes. Accordingly, the following research questions are proposed:

RQ1: To what extent does targeted metacognitive scaffolding improve primary school students' critical thinking performance in GenAI-supported collaborative learning?.

RQ2: How does metacognitive scaffolding influence the depth of students' cognitive engagement during collaborative interaction with peers and GenAI?.

RQ3: What observable behavioral sequences and epistemic network characteristics emerge within the Learner–Peer–GenAI interaction under scaffolded conditions?

3. Design of metacognitive scaffolds for GenAI-mediated collaboration

3.1. Framework design

The scaffolding framework in this study is grounded in a ternary interaction model (Learner–Peer–GenAI) (as shown in Figure 1), specifically engineered to address the epistemic shifts inherent in AI-supported collaborative environments. Unlike traditional scaffolds that primarily focus on interpersonal coordination, this design targets the regulation of "cognitive offloading" – a phenomenon where learners may uncritically internalize GenAI's syntactically fluent but potentially erroneous outputs. By integrating metacognitive prompts into the human–AI interface, the scaffold functions as a regulatory mechanism that transforms GenAI from a passive information provider into a provocative cognitive agent. This approach necessitates active verifica-

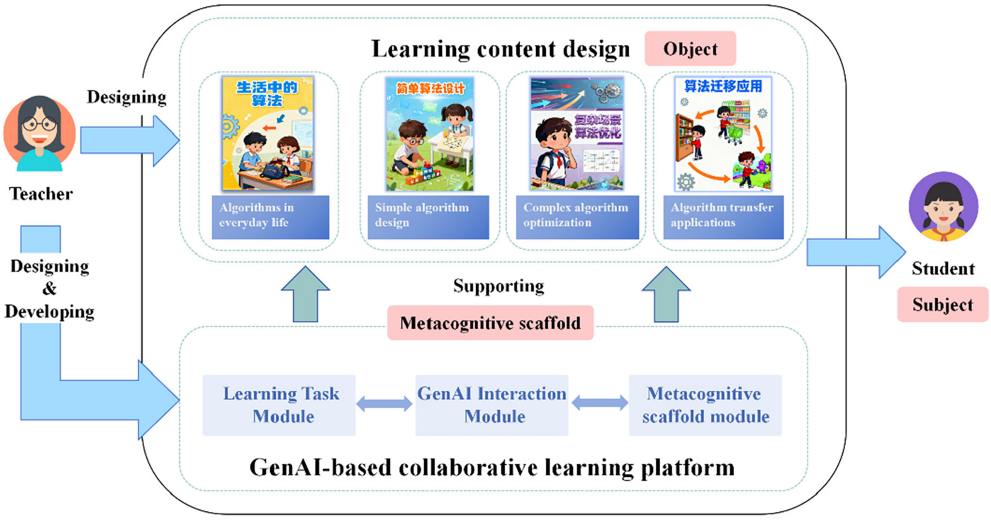


Figure 1. Design of metacognitive scaffolds for GenAI-mediated collaboration.

tion and logical deconstruction, ensuring that learners maintain epistemic agency while navigating the "pseudo-authority" of algorithmic outputs.

3.2. Learning task design

To establish the theoretical necessity of GenAI within the research design, the learning tasks were operationalized around the construct of "Epistemic Friction." These tasks involve complex problem-solving scenarios where GenAI is strategically configured to generate "plausible but suboptimal" outputs characterized by clandestine logical inconsistencies (as shown in Table 1). Consequently, the pedagogical focus shifts from rote task completion to "Collaborative AI-Auditing." Within this framework, students are compelled to maintain heightened metacognitive vigilance to detect and reconcile algorithmic "hallucinations" or logical lacunae. This structural design ensures that the unique affordances and constraints of GenAI are functionally intertwined with the learning process, positioning the technology as an indispensable catalyst for the targeted metacognitive interventions.

3.3. Design framework for metacognitive scaffolding

Sixth-grade learners (typically aged 11–12) are at a transitional stage from concrete operational to formal operational thinking. Although they have begun to show foundational metacognitive abilities, they still benefit from structured guidance when engaging with GenAI outputs. For this reason, the metacognitive scaffolding framework in the present study was intentionally designed as an age-appropriate support for sixth-grade students. The metacognitive scaffolds are organized into three interconnected regulatory phases, each aligned with the dynamics of AI-mediated inquiry (Ataş & Yıldırım, 2025). In the Strategic Planning phase, the scaffolds are intended to reduce observable dependence on initial AI-generated frames by prompting groups to articulate "interrogation boundaries" and explicit verification criteria before engaging with AI. In the Active Monitoring phase, the intervention deploys structured logical-sorting prompts that require students to cross-verify claims and apply counterfactual reasoning to the AI's output, shifting engagement from surface-level retrieval to adversarial, evidence-oriented reasoning. In the Evaluative Reflection phase, the scaffolds target the refinement of human–AI interaction patterns by directing learners to appraise their prompt-construction strategies and to diagnose episodes of cognitive inertia during collaboration (as shown in Table 2). Collectively, these phases cultivate a recursive metacognitive routine that trains learners not only to interrogate GenAI outputs, but also to monitor and iteratively improve their own reasoning throughout AI-mediated tasks.

3.4. System platform

To embed metacognitive scaffolds seamlessly into GenAI-supported collaboration, we developed a custom web-based platform (Figure 2) that communicated with the Doubao large language model via API. The

Table 1. Design of learning tasks integrated with metacognitive scaffolding in GenAI-supported collaboration.

Learning phase	Task goal & role of GenAI	Potential cognitive risks	Focus of metacognitive Scaffolding	Example scaffold prompts/ activities
Planning	Algorithmic clarification: Define exploratory tasks and anticipate the solution framework; GenAI provides an initial solution proposal.	Tendency to accept the first AI-generated solution without scrutiny.	Epistemic standard setting: Guide students to establish evaluation criteria and reasoning pathways before engaging with AI outputs.	"Before reviewing the AI-generated solution, what assumptions do we expect it to make? What evidence or reasoning should be provided?"
Monitoring	Adversarial verification: Critically examine and revise AI-generated code or solutions during task execution.	Cognitive offloading: AI's fluent explanations may obscure logical gaps, leading to superficial understanding.	Epistemic reasoning prompts: Encourage logical consistency checks and counterfactual reasoning.	"Please identify the assumptions underlying the AI's input and output. If one condition changes, does the conclusion still hold?"
Reflection	Algorithmic internalization: Evaluate outcomes and reflect on human–AI interaction strategies.	Cognitive inertia: Overreliance on task completion results may suppress reflection on reasoning biases.	Strategy transfer and optimization: Support learners in extracting generalized strategies from prior AI-supported inquiry.	"How did our prompting strategies influence the AI's responses? How can we adjust them to avoid hallucinations or reasoning flaws next time?"

Table 2. A design framework for metacognitive scaffolding tailored for GenAI contexts.

Phase	Scaffolding type	Core objective	GenAI-specific interaction strategy
Planning	Procedural Scaffolding	To establish inquiry boundaries and discourage uncritical acceptance of AI-generated outputs.	1. Guide students to design targeted "stress-test" question sets; 2. Use AI to decompose tasks while requiring students to manually filter and refine results based on pre-set criteria.
Monitoring	Monitoring Scaffolding	To activate Epistemic Vigilance and perform real-time algorithmic auditing.	1. Mandate GenAI to provide source citations for factual verification; 2. Utilize AI as a "Devil's Advocate" to challenge the logical integrity of students" or peers" arguments (Adversarial Inquiry).
Reflection	Reflective Scaffolding	To optimize the "Human-AI-Peer" triadic collaborative synergy.	1. Prompt groups to ask GenAI: "Comparing our initial plan with the final conclusion, where did cognitive biases occur?"; 2. Use AI to retrieve diverse "exemplars" for comparative analysis to identify students" own cognitive blind spots.

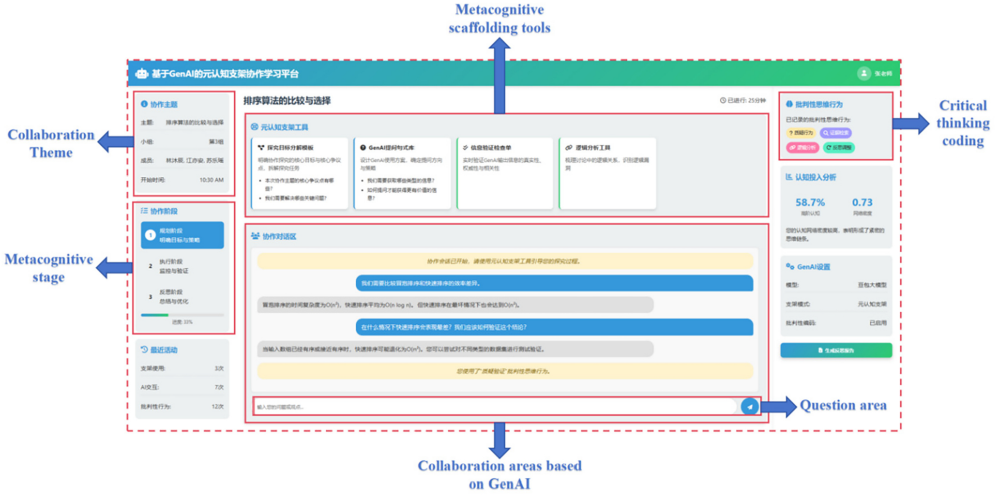


Figure 2. Metacognitive scaffolding collaborative platform based on GenAI.

platform was designed not merely as an access interface to GenAI, but as a regulation-oriented learning environment in which the scaffold was integrated into the interaction process itself. Its core component, the Embedded Metacognitive Toolbox, enabled real-time, context-sensitive scaffolding while preserving the natural flow of learner–peer–GenAI collaboration.

The platform served four main functions (Figure 3). First, it supported multi-channel interaction among learners, peers, and GenAI within a shared task environment. Second, it logged students' prompts, peer exchanges, and AI-response use in real time. Third, it continuously analysed behavioral patterns associated with possible cognitive offloading, such as copying AI-generated output within 15 s without modification, producing three consecutive turns directed only to AI, or proceeding through five turns without any questioning behavior. Fourth, when such indicators reached a predefined threshold, the scaffolding engine retrieved a matched prompt from the metacognitive prompt library and delivered it through a brief sidebar pop-up. In this sense, the platform operationalized a closed regulatory loop: interaction logging → behavioral analysis → offloading-risk detection → scaffold retrieval → prompt delivery → renewed learner regulation.

Importantly, the form of regulation in this study was system-initiated and embedded, rather than teacher-delivered. All metacognitive prompts were hard-coded into the platform and automatically triggered by behavioral indicators. Teachers did not provide the prompts, interpret them during task execution, or intervene in the timing of scaffold delivery; their role was limited to classroom organization and procedural support. This design ensured that the intervention remained consistent across groups and sessions and enhanced the replicability of the quasi-experimental procedure.

To ensure developmental appropriateness for sixth-grade learners, the prompts were distributed across the planning, monitoring, and reflection phases and were phrased in a directive yet open-ended form so as to support questioning, verification, and reflection without imposing excessive procedural demands. Once triggered, the prompts appeared in a non-intrusive sidebar rather than interrupting the main task interface.

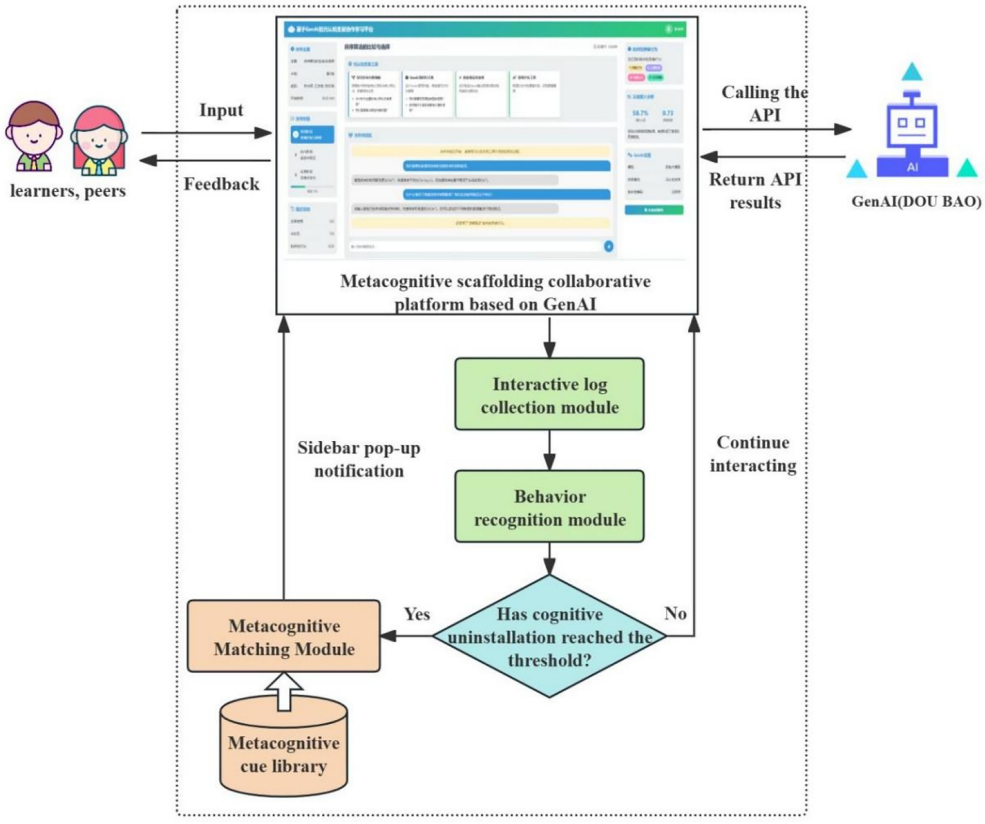


Figure 3. Metacognitive scaffold collaboration mechanism based on GenAI.

In this way, the platform was designed not simply to provide assistance, but to reorient learners' epistemic engagement with AI outputs and reduce the likelihood of uncritical dependence during collaborative inquiry.

4. Research methods

4.1. Participants

To ensure the scientific rigor and comparability of the experimental results, this study selected 85 students from two parallel sixth-grade classes in a certain XX Primary School as the research subjects. All students had basic information technology operation abilities (e.g. being able to independently log in to the learning platform and input text), which laid a technical foundation for the smooth implementation of the experiment. According to the principle of random assignment, 42 students in Class 1 (22 boys and 20 girls) were set as the experimental group, and 43 students in Class 2 (22 boys and 21 girls) were set as the control group. Both groups were taught by the same instructor, an expert in information technology education with over a decade of experience, ensuring instructional consistency. Before the formal intervention, this study administered a pre-test to examine the differences between the two groups in terms of critical thinking level and cognitive engagement level to verify the homogeneity between the groups. The results showed that there were no significant differences between the two groups of students in the total score of critical thinking and its three core dimensions (questioning and verification, logical reasoning, and reflection and adjustment), indicating that the experimental group and the control group were at the same baseline level before the intervention (as shown in Table 5), which provided reliable conditions for the subsequent comparison of the intervention effects. To maintain ethical standards, informed consent was obtained from all participants and their guardians, and the study was conducted in accordance with the institution's research ethics guidelines. In addition, because the study relied on a platform that recorded learner interaction data, several privacy safeguards were implemented. Only task-relevant learning-process data were collected for analysis, and no

unnecessary personally sensitive information was used. All student records were anonymized prior to export and analysis by replacing names with coded identifiers. Access to dialogue logs and platform records was restricted to the research team, and all excerpts reported in the manuscript were presented in de-identified form to protect participants' privacy.

4.2. Experimental procedures

This eight-week quasi-experimental study was designed to position GenAI as an indispensable, non-neutral variable rather than a background tool. To introduce "epistemic friction," the collaborative activities were constructed as algorithm-auditing tasks. Through tailored prompt engineering, the GenAI was configured to generate initial solutions that appeared plausible yet contained logically flawed reasoning or hallucinated claims, thereby requiring students to subject the outputs to rigorous scrutiny. The experimental group engaged with an embedded metacognitive scaffolding system optimized for human–AI collaboration. During the planning phase, the scaffolds prompted students to specify the division of labor between human reasoning and AI assistance. During the monitoring phase, the system tracked real-time discourse patterns and, when indicators of cognitive offloading emerged (e.g. direct copy-pasting of AI outputs), automatically issued adversarial prompts (e.g. "Does this AI-generated logic hold under boundary conditions?"). By contrast, the control group completed the same GenAI-supported collaborative tasks without targeted metacognitive supports and received only standard task instructions.

To eliminate confounding variables stemming from technical disparity, both groups utilized an identical GenAI platform (Doubao) throughout the intervention. Following each session, students were mandated to submit a multi-modal portfolio of learning artifacts. This included the "Algorithm Exploration Handbook" – a structured repository of procedural diagrams, verification logs, and optimization schemata – alongside verbatim transcripts of collaborative dialogues for subsequent process-tracing and behavioral coding. The eight-week intervention was synchronized with the standard primary school academic calendar to preserve ecological validity. The longitudinal implementation workflow is systematically delineated in Figure 4.

To enhance procedural rigor and experimental control, the eight-week cycle was structured into five sequential stages. Week 1 served as the preparation stage, during which students received standardized training on the GenAI tool and completed within-group role allocation to establish the technical and organizational foundation. During this orientation session, both the experimental and control groups received identical instruction on the basic operation of the GenAI platform (Doubao), including prompt submission, interpretation of AI-generated responses, and documentation of interaction processes. Importantly, none of the participants had previously engaged in structured classroom activities involving GenAI

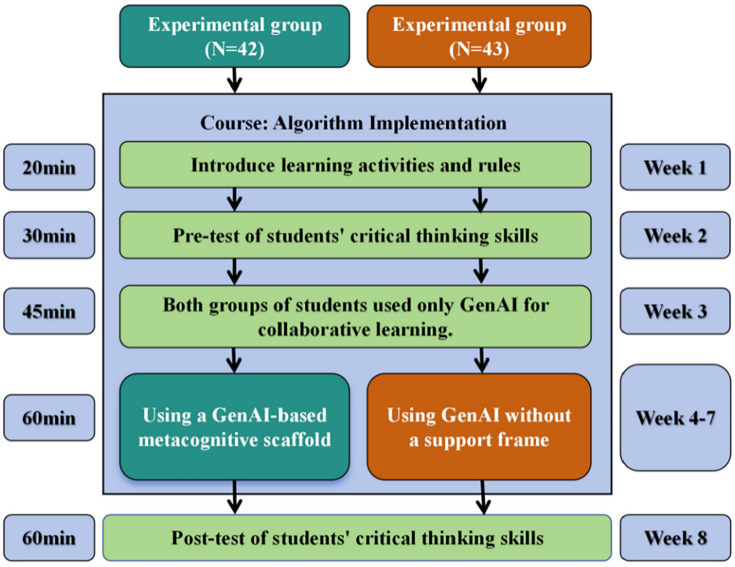


Figure 4. Experimental procedure.

tools prior to the study. This preparatory training therefore ensured a comparable level of operational AI literacy across the two groups before the formal collaborative tasks began. Week 2 comprised the pre-test stage: students completed a critical-thinking pre-test to establish baseline performance and to assess between-group equivalence. Week 3 functioned as the baseline period, in which both groups completed the collaborative tasks with GenAI but without metacognitive scaffolds, allowing observation of natural collaborative performance and establishing a benchmark for comparison. This baseline stage further allowed the researchers to observe students' natural interaction patterns with GenAI prior to the scaffolding intervention.

Weeks 4–7 constituted the intervention period. The experimental group systematically implemented three scaffold types – planning, monitoring, and reflection – to support structured collaboration and submitted records of scaffold use. The control group continued GenAI-supported collaboration without scaffolds and submitted routine learning logs. Week 8 served as the transfer and post-test stage. All scaffolds were removed to evaluate students' transfer performance, after which a critical-thinking post-test was administered and all process data were consolidated to support comprehensive analyses of intervention effects.

To prevent contamination of scaffold-related process data, only interactions recorded during the intervention period (Weeks 4–7) were included in process-level analyses (e.g. Epistemic Network Analysis and Lag Sequential Analysis). Baseline interactions from Week 3 – when both groups used GenAI without scaffolds – were excluded to preserve the internal validity of scaffold-linked behavioral inferences.

4.3. Data collection

4.3.1. Critical thinking scale

To rigorously quantify students' critical thinking, this study utilized a revised version of the "Critical Thinking Ability Assessment Scale for Primary School Students." Grounded in Ennis' critical thinking theory and Facione's Delphi Report framework, the scale underwent context-specific adaptation to align with the "algorithmic learning" paradigm in primary education, thereby ensuring robust content and ecological validity. The instrument comprises 20 items distributed across three dimensions: Evidence Verification (7 items), Logical Reasoning (7 items), and Reflective Adjustment (6 items). Responses are captured via a 5-point Likert scale, ranging from 1 ("completely unable to perform") to 5 ("consistently able to perform"), yielding a total score between 20 and 100. This psychometric approach is specifically optimized for primary learners, facilitating the collection of stable, reliable data while maintaining cognitive accessibility.

A pilot study ($N = 30$) was conducted to evaluate the internal consistency and construct validity of the scale. The instrument yielded a robust overall Cronbach's α of 0.82, with subscale coefficients of 0.78 for Evidence Verification, 0.76 for Logical Reasoning, and \$0.79\$ for Reflective Adjustment, indicating satisfactory-to-high reliability. To interrogate the scale's structural integrity, an Exploratory Factor Analysis (EFA) was performed using principal component analysis. Three factors with eigenvalues exceeding 1.0 were extracted, collectively accounting for 68.3% of the total variance. Factor loadings predominantly exceeded 0.60, with items consistently aligning with their a priori theoretical constructs. These findings substantiate the tri-dimensional architecture of the scale, confirming its psychometric suitability for assessing critical thinking in primary education contexts.

4.3.2. Solo cognitive level coding system

To evaluate the cognitive depth of collaborative discourse, this study developed a coding scheme based on Biggs and Collis' SOLO Taxonomy, specifically adapted for GenAI-supported environments (see Table 3). To ensure the scientific rigor and objectivity of the qualitative analysis, two systematically trained researchers independently coded the entire corpus of dialogue transcripts. Prior to the formal coding phase, an inter-rater reliability test was conducted on a randomly selected subset (10%) of the data. The resulting Cohen's Kappa coefficient was 0.86, indicating high inter-rater agreement and confirming the reliability of the coding framework. Discrepancies identified during this phase were resolved through iterative discussion until a consensus was reached, ensuring the integrity of the subsequent process-level analysis.

Crucially, to isolate student-generated cognitive effort from the highly structured outputs of GenAI, we implemented a strict 'Transformation Rule.' Any utterance consisting solely of copy-pasted AI content was

Table 3. SOLO cognitive level coding system in collaborative learning.

SOLO level	Encoding	Core features
Prestructural	CP	The discussion content is irrelevant to the exploration objective, or there are serious misunderstandings about GenAI or peers' viewpoints, making it impossible to form an effective exploration idea.
Unistructural	CU	Only focusing on a single information point of GenAI or peers, without logical extension and in-depth thinking, and only conducting simple repetition or agreement.
Multistructural	CM	Attention is paid to multiple information points, but no logical associations are established among the information. Instead, only fragmented listings are made, lacking integration and analysis.
Relational structure	CR	Integrate multiple information points, establish logical connections, and form a coherent argument or solution. It is also capable of analyzing relationships such as causality and parallelism among the information.
Extended structure	CE	Generating abstract laws or general frameworks based on logical integration enables the transfer of learned knowledge to new scenarios, or actively questioning the limitations of GenAI or peers' viewpoints and proposing innovative suggestions.

automatically coded at the Unistructural (CU) or Multistructural (CM) level, reflecting the lack of internal cognitive processing (i.e. cognitive offloading). Higher-level codes (Relational/CR and Extended/CE) were assigned only when students demonstrated explicit transformation of AI outputs – such as critiquing logical flaws, cross-verifying with external evidence, or synthesizing AI suggestions with peer perspectives. This approach ensures that the SOLO scores reflect the learners' epistemic agency rather than the generative capability of the AI.

4.3.3. Coding system for critical thinking behaviors

To address the unique dynamics of GenAI-mediated environments, we extended the traditional critical thinking behavioral coding scheme by incorporating AI-specific interaction dimensions (QB.AI, ER.AI, and LA.AI) (Table 4). Specifically, "Questioning Behavior toward AI" (QB.AI) was defined as active skepticism regarding the veracity and logical coherence of AI outputs. "Evidence Retrieval toward AI" (ER.AI) involved cross-referencing AI-generated claims with external authoritative sources. "Logical Reconstruction toward AI" (LA.AI) referred to the corrective actions taken by students to refine flawed AI algorithms. This refined coding scheme enables the analysis of observable dependence-related discourse behaviors and heuristic trust-related interaction patterns toward AI-generated outputs, thereby clarifying how scaffolding may foster critical agency in AI-enhanced settings. This system provides clear node definitions for subsequent cognitive network analysis, thus ensuring the structuring and operability of the analysis framework. In the specific implementation process, two researchers independently coded all the collaborative dialogue texts, and the obtained Kappa coefficient was 0.83, indicating that the coding results have good consistency, which lays a reliable foundation for further data analysis.

Table 4. Coding system for critical thinking behaviors in algorithmic collaborative learning.

Behavior types	Encoding	Behavior description
Information Borrowing	IB	Directly repeating the content of GenAI or peers without any thinking, analysis, or processing, only performing simple information transfer.
Simple Agreement	SA	Agreeing with others' viewpoints or the output results of GenAI without basis, without conducting logical analysis or evidence verification, and only expressing an attitude of agreement
Questioning Behavior	QB	Raise questions about the rationality of peer opinions, the sufficiency of evidence, the feasibility of algorithms, etc., including clear objects of doubt and problem directions.
Questioning Behavior toward AI	QB.AI	Raise questions about the rationality of GenAI, the sufficiency of evidence, the feasibility of algorithms, etc., including clear objects of doubt and problem directions.
Evidence Retrieval	ER	Actively seek evidence, data, or cases to support or refute viewpoints and verify the feasibility of algorithms through channels such as GenAI, textbooks, and online resources.
Evidence Retrieval toward AI	ER.AI	Cross-verify AI outputs by consulting external sources.
Logical Analysis	LA	Sort out the logical relationships such as causality and parallelism between viewpoints and algorithm steps, and identify logical loopholes, algorithmic defects, or reasoning contradictions.
Logical Analysis toward AI	LA.AI	Algorithm corrections initiated after identifying AI vulnerabilities
Evaluation Judgment	EJ	Evaluate the information quality, the rationality of viewpoints, and the advantages and disadvantages of algorithms, and provide clear judgment bases and evaluation criteria.
Reflection Adjustment	RA	Examine the cognitive biases, strategic deficiencies, or algorithmic issues in the collaborative process, and adjust the exploration strategies, GenAI usage methods, or algorithm design schemes.
Consensus Construction	CC	Integrate diverse perspectives through critical analysis to form consistent conclusions, solutions, or algorithm designs, and clarify the core content and formation process of the consensus.

Table 5. Pretest of critical thinking Levels in the experimental group and the control group.

	Groups (mean ± standard deviation)			
	Experimental group (n = 42)	Control group (n = 43)	t	p
questioning and verification	20.84 ± 3.96	21.12 ± 4.02	−0.31	0.756
Logical reasoning	21.37 ± 3.88	21.06 ± 3.91	0.36	0.721
Reflection and adjustment	18.95 ± 3.41	19.14 ± 3.36	−0.27	0.789

*p < 0.05; **p < 0.01.

4.4. Data analysis

To address the three research questions in a comprehensive and systematic manner, this study adopted a complementary multi-method analytic strategy to clarify how metacognitive scaffolding reconfigures human–AI interaction dynamics. First, overall changes in critical thinking were examined using descriptive statistics and independent-samples t tests in SPSS 26.0, testing between-group differences in total scores and each critical-thinking dimension. Second, to assess cognitive engagement during collaborative dialogue, the full transcript corpus was coded using the SOLO taxonomy. Proportions across SOLO levels were computed for each group, and chi-square tests were conducted to evaluate between-group differences in level distributions. Third, to characterize structural shifts in critical thinking within interaction, Epistemic Network Analysis (ENA 1.6.0) was applied. Treating the critical-thinking behavioral codes (Table 3) as network nodes, we compared groups on connection strength, network density, and centroid positions to interpret differences in cognitive organization. Finally, to trace the temporal dynamics of critical-thinking behaviors, Lag Sequential Analysis (LSA) was conducted in GSEQ 5.1. Statistically significant transition paths (Z > 1.96) were identified and contrasted across groups to delineate distinct behavioral sequences and interactional regularities.

5. Result analysis

5.1. Differences in critical thinking levels

Independent-samples t tests were conducted to examine differences in critical thinking levels between the experimental and control groups before and after the intervention. As shown in Table 5, no significant differences were found between the two groups in the pre-test across any of the three dimensions – questioning and verification, logical reasoning, and reflection and adjustment – or in the total critical thinking score (all ps > 0.70). These results indicate satisfactory baseline equivalence and suggest that the two groups were comparable prior to the intervention.

Post-test results, presented in Table 6, revealed significant differences between the experimental and control groups across all dimensions of critical thinking. Students in the experimental group outperformed those in the control group in questioning and verification (t = 4.86, p < 0.01), logical reasoning (t = 3.58, p < 0.01), and reflection and adjustment (t = 3.24, p < 0.01). Notably, the most pronounced difference was found in the questioning and verification dimension (t = 4.86, p < 0.01), whereas the differences in logical reasoning and reflection and adjustment were moderate in magnitude.

This non-uniform pattern of improvement suggests that the intervention did not simply enhance students' general critical thinking ability. Rather, the substantial gain in questioning and verification indicates that the metacognitive scaffolding was particularly effective in activating students' systematic doubt, evidence checking, and evaluative judgment toward GenAI-generated content. In the GenAI-supported collaborative learning context, students in the experimental group were more likely to critically interrogate AI outputs, verify their accuracy, and reconsider their validity, instead of accepting them as authoritative

Table 6. Post-test of critical thinking skills in the experimental group and the control group.

	Groups (M ± SD)				
	Experimental group (n = 42)	Control group (n = 43)	t	p	Cohen's d [95% CI]
Question and verification	26.48 ± 3.72	22.03 ± 4.01	4.86	<0.01**	1.05 [0.60, 1.51]
Logical reasoning	24.19 ± 3.65	22.11 ± 3.84	3.58	<0.01**	0.78 [0.34, 1.22]
Reflection and adjustment	22.06 ± 3.18	20.01 ± 3.27	3.24	<0.01**	0.70 [0.26, 1.14]

**p < 0.01.

sources.Taken together, these findings suggest that the effectiveness of the metacognitive scaffolding lies not in promoting undifferentiated critical thinking growth, but in fostering domain-specific epistemic vigilance toward algorithmically generated information.

5.2. Hierarchical analysis of cognitive engagement

To further investigate the impact of the intervention on cognitive engagement, collaborative discourse was systematically analyzed using the SOLO taxonomy at the utterance level. A total of 1223 discourse turns were discretized and coded, comprising 612 utterances from the experimental group and 611 from the control group. This nearly identical distribution of conversational turns ensures a balanced discourse corpus and reduces the likelihood that differences in subsequent process analyses (e.g. ENA and LSA) are attributable to unequal discussion volume between the two groups. This nearly identical distribution of conversational turns ensures a balanced corpus for inter-group comparison. Table 7 summarizes the descriptive distribution of cognitive engagement levels, providing a foundation for evaluating the complexity of knowledge co-construction across both conditions.

The chi-square test indicated a significant between-group difference in the distribution of SOLO levels, $\chi^2(4, N = 1223) = 118.63$, $p < 0.001$, with a moderate effect size (Cramer's $V = 0.31$). This pattern suggests a systematic association between instructional condition and the cognitive level of students' discourse.

As shown in Table 7, the experimental group exhibited a substantially higher proportion of high-level cognitive engagement than the control group. Specifically, 49.7% of the experimental group's utterances were coded at the relational or extended abstract levels, compared with 27.3% in the control group. By contrast, the control group produced a larger share of low-level utterances, particularly at the prestructural and unistructural levels, indicating greater reliance on fragmented or surface-level responses.

Between-group differences were most pronounced at the highest SOLO level. Extended abstract utterances constituted 17.5% of the experimental group's discourse, versus 6.2% in the control group. These utterances typically reflected abstraction and generalization, critical reinterpretation of information, or reflective reasoning that extended beyond the immediate task context. The markedly higher prevalence of such responses in the experimental group suggests that metacognitive scaffolding in a GenAI-supported collaborative environment promoted higher-order conceptualization and reflective thinking. More broadly, the results imply that the scaffolds functioned not only as a catalyst for higher-order cognition but also as a regulatory mechanism that helped manage cognitive demands in human–AI collaboration – redirecting learners' limited cognitive resources from peripheral coordination toward deep logical construction and algorithm auditing.

5.3. Development trajectory of critical thinking

In this study, the ENA 1.6.0 WebTool was employed to conduct cognitive network analysis. Critical thinking behavior codes were used as network nodes to construct the cognitive networks of the experimental group and the control group. It should be noted that the ENA comparison focuses on normalized co-occurrence patterns among coded behaviors rather than raw utterance counts. Consequently, the observed differences between groups represent shifts in the proportional organization of epistemic actions within discourse rather than simple increases in conversational output. By analyzing the node co-occurrence intensity, network density, centroid distribution, and its 95% confidence interval, this study aimed to reveal the dynamic association patterns of critical thinking behaviors in the two groups. The results revealed a clear divergence in how students cognitively positioned GenAI during collaboration. In the experimental group, AI-oriented

Table 7. Distribution of cognitive engagement Levels based on SOLO taxonomy (utterance count).

SOLO level	Experimental group ($n = 612$)	Control group ($n = 611$)	Total ($N = 1223$)
Prestructural	52 (8.5%)	104 (17.0%)	156 (12.8%)
Unistructural	118 (19.3%)	167 (27.3%)	285 (23.3%)
Multistructural	138 (22.5%)	173 (28.3%)	311 (25.4%)
Relational structure	197 (32.2%)	129 (21.1%)	326 (26.7%)
Extended structure	107 (17.5%)	38 (6.2%)	145 (11.9%)

Note. χ^2 (4, $N = 1223$) = 118.63, $p < 0.001$, Cramer's $V = 0.31$, 95% CI [0.25, 0.36].

high-order behaviors – Questioning Behavior toward AI (QB.AI), Evidence Retrieval toward AI (ER.AI), and Logical Analysis toward AI (LA.AI) – occurred far more frequently than in the control group. This pattern suggests that the metacognitive scaffolding helped reposition GenAI from an "answer authority" to an object of inquiry and auditing, thereby fostering a planning–monitoring–reflection regulatory loop associated with higher-order cognitive processing. Rather than merely accepting AI-generated responses, learners engaged in questioning, evidence verification, logical reconstruction, evaluative judgment, and reflective adjustment (Shekh-Abed, 2025; Yazici & Dogan, 2025).

The cognitive network for the experimental group (Figure 5) displayed a high-density, strongly synergistic configuration. Its core was no longer driven primarily by generic questioning or reasoning; instead, it was structured around a GenAI-auditing triad anchored by the sequence QB.AI → ER.AI → LA.AI. Functionally, learners first articulated targeted skepticism toward AI outputs (QB.AI), then cross-validated those outputs using external sources (ER.AI), and subsequently reconstructed the underlying logic and revised algorithmic judgments when vulnerabilities or inconsistencies were detected (LA.AI). Importantly, this core cluster radiated outward to evaluative and regulatory nodes (EJ, RA) and converged on CC, forming a coherent epistemic progression in which verification and reconstruction became routinized rather than episodic. This structure offers process-level evidence that the scaffolding's planning prompts prime students to establish epistemic standards prior to AI use, monitoring prompts sustain real-time evidence checking, and reflection prompts consolidate reconstruction strategies into transferable self-regulation – collectively producing a stable pathway for higher-order thinking in human–AI collaboration.

In contrast, the cognitive network of the control group manifests a centralized, low-association structure, characterized by a significantly lower network density compared to the experimental group (Figure 6). Interaction patterns were heavily skewed toward Information Borrowing (IB) and Simple Agreement (SA), indicating that, despite the same task environment designed to introduce epistemic friction, students in the unscaffolded condition tended to adopt low-cost interaction strategies – treating GenAI as a convenient information source and defaulting to rapid acceptance rather than systematic scrutiny. From a network-analytic perspective, this concentration underscores a low-density architecture anchored in an "information acquisition–rapid endorsement" loop, wherein high-order cognitive nodes (e.g. questioning, evidence retrieval, logical analysis, and reflective adjustment) remain peripheral and sparsely connected. Consequently, these findings are consistent with a pattern of AI-related cognitive offloading, whereby learners

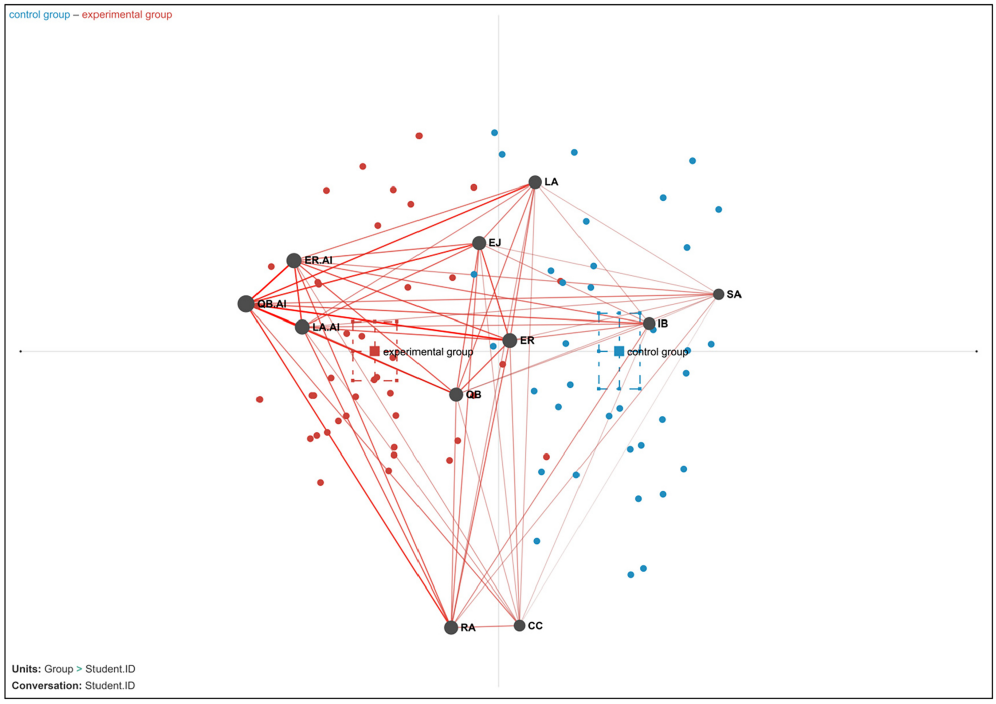


Figure 5. Cognitive network diagram of the experimental group. The network was visualized using the Fruchterman-Reingold force-directed layout. Edge thickness indicates normalized co-occurrence strength.

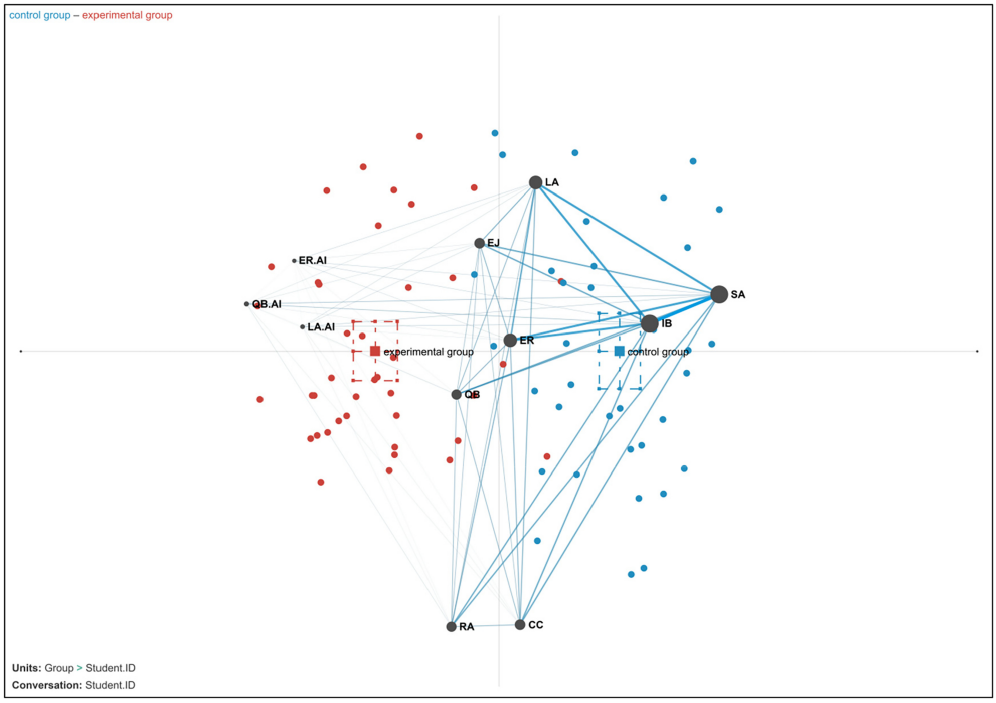


Figure 6. Cognitive network diagram of the control group. The network was visualized using the Fruchterman-Reingold force-directed layout. Edge thickness indicates normalized co-occurrence strength.

increasingly shift epistemic responsibility for verification and evaluative judgment to AI-generated outputs, allowing perceived fluency and authority to bypass systematic reasoning and reflective reconstruction.

To more intuitively present the structural differences between the two groups, this study further conducted a subtraction comparison of the cognitive networks of the two groups (Figure 7). The results showed that the experimental group had a significant co-occurrence intensity advantage in the QB.AI–ER.AI–LA.AI

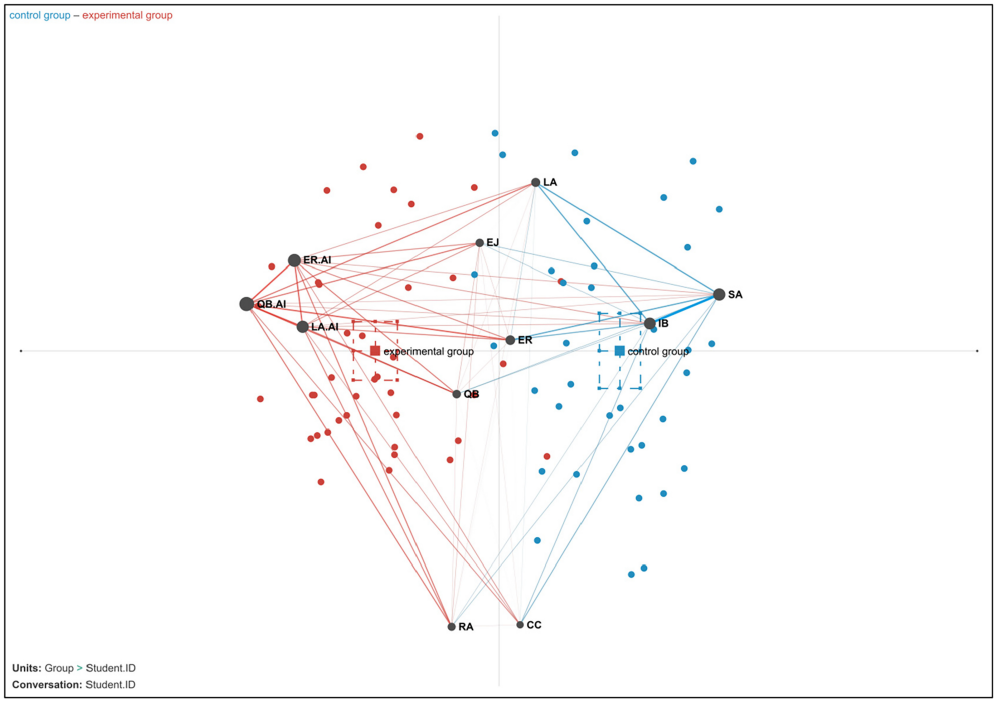


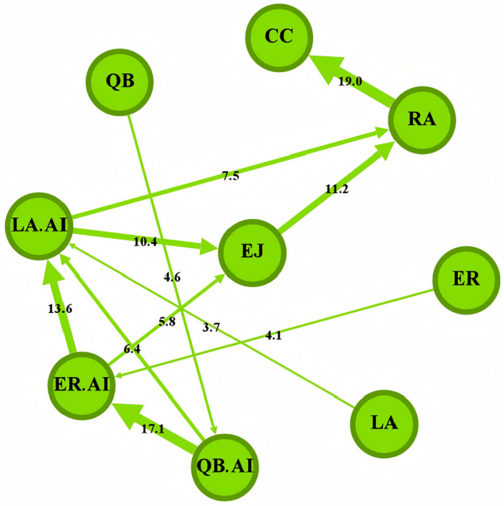
Figure 7. Subtraction maps of cognitive networks between the experimental group and the control group.

core cluster and its associated regions with EJ, RA, and CC. In contrast, the control group only exhibited relatively prominent but still limited co-occurrence characteristics in the IB–SA connection region. This structural difference indicates that metacognitive scaffolding does not merely increase "more discussion," but specifically reconfigures the distribution and connectivity of AI-targeted critical behaviors. By amplifying QB.AI, ER.AI, and LA.AI, the scaffolding operationalizes the planning–monitoring–reflection cycle as an epistemic control system that prevents IB/SA-dominated superficiality and enables sustained higher-order cognition in GenAI-supported collaborative learning.

5.4. Critical thinking behavior sequence

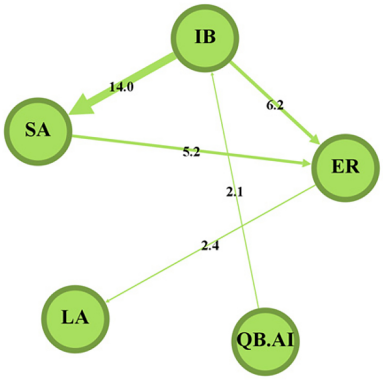
To examine how metacognitive scaffolding reshaped the temporal dynamics of students' critical thinking in GenAI-supported collaboration, we performed lag sequential analysis (LSA) on the coded discourse sequences using GSEQ 5.1. Statistically reliable transitions were identified via adjusted residual Z values, with $Z > 1.96$ indicating a significant lagged association. Figures 8 and 9 depict the significant transition patterns for the experimental and control groups, respectively; arrow direction denotes lag order, and line thickness indicates transition salience (i.e. Z magnitude).

As shown in Table 8 and Figure 8, the experimental group demonstrated a highly organized, higher-order sequential structure characterized by an "AI-auditing progression," rather than a superficial information loop. The dominant pathway reflected AI-targeted epistemic vigilance: QB.AI → ER.AI ($Z = 17.1$) and ER.AI → LA.AI



Code	code name
QB	Questioning Behavior
QB.AI	Questioning Behavior toward AI
ER	Evidence Retrieval
ER.AI	Evidence Retrieval toward AI
LA	Logical Analysis
LA.AI	Logical Analysis toward AI
EJ	Evaluation Judgment
RA	Reflection Adjustment
CC	Consensus Construction

Figure 8. Analysis of lagged behavioral sequences in the experimental group.



Code	code name
IB	Information Borrowing
SA	Simple Agreement
QB.AI	Questioning Behavior toward AI
ER	Evidence Retrieval
LA	Logical Analysis

Figure 9. Analysis of lagged behavior sequences in the control group.

Table 8. Residual transformation results of the experimental group.

Behavioral coding	IB	SA	QB	QB.AI	ER	ER.AI	LA	LA.AI	EJ	RA	CC
IB	−1.6	−1.9	1.1	0.8	0.9	0.6	0.7	0.5	0.3	0.2	0.1
SA	−1.8	−1.5	1.2	0.9	1	0.7	0.8	0.6	0.4	0.3	0.2
QB	−1.9	−2	−1.1	4.6*	1.7	1.4	1.6	1.5	1.2	1	0.8
QB.AI	−2.1	−2	−1.3	−1.2	1.8	17.1*	1.6	6.4*	1.5	1.9	1.4
ER	−2	−1.8	1.2	1.5	−1.2	4.1*	1.7	1.6	1.4	1.3	1.1
ER.AI	−2.2	−2.1	1	1.4	1.6	−1.3	1.8	13.6*	5.8*	1.9	1.2
LA	−2.1	−1.9	1.1	1.3	1.5	1.6	−1	3.7*	1.8	1.6	1.4
LA.AI	−2	−1.8	1.2	1.5	1.6	1.8	1.4	−1.1	10.4*	7.5*	1.8
EJ	−1.9	−1.7	1.1	1.3	1.4	1.6	1.5	1.7	−1	11.2*	1.6
RA	−2.2	−2	1	2.3*	1.3	1.5	1.4	1.6	1.8	−1.2	19.0*
CC	−1.6	−1.5	1.5	1.8	1.4	1.6	1.3	1.7	1.9	1.8	−0.9

*$p < 0.05$.

($Z = 13.6$), suggesting that skepticism toward AI outputs consistently prompted external cross-verification and then led to algorithm-focused correction and reconstruction. This AI-oriented chain further extended into evaluative and regulatory phases, as indicated by LA.AI → EJ ($Z = 10.4$) and EJ → RA ($Z = 11.2$), culminating in RA → CC ($Z = 19.0$). Collectively, once AI vulnerabilities were detected and addressed, students were more likely to appraise solution quality, recalibrate strategies, and consolidate consensus through an explicit reasoning cycle – yielding a coherent higher-order trajectory from interrogation to convergence. It is also noteworthy that ER/ER.AI behaviors in the experimental group frequently appeared as clusters of consecutive discourse turns involving information searching, AI re-querying, and peer negotiation. Such multi-turn interactional clusters function as a proxy for sustained verification activity, suggesting that evidence retrieval represented a substantive reasoning phase rather than a momentary transition within the dialogue.

Significantly, the experimental group's sequence exhibited "bridging transitions" that elucidate how high-order cognitive chains were initiated and sustained within the GenAI context. Specifically, the transition QB → QB.AI ($Z = 4.6$) suggests that general questioning was systematically redirected toward the critical interrogation of AI outputs. Similarly, ER → ER.AI ($Z = 4.1$) indicates that broader evidence-seeking frequently escalated into targeted external verification of AI-generated claims. Furthermore, LA → LA.AI ($Z = 3.7$) demonstrates that logical analysis consistently transmuted into concrete algorithmic revision upon the detection of flaws in AI reasoning. Finally, the RA → QB.AI ($Z = 2.3$) pathway instantiates a recursive regulatory mechanism: reflective adjustment frequently re-triggered AI-directed questioning. This pattern corroborates the operation of a planning–monitoring–reflection loop as a continuous, iterative epistemic control system, rather than a transient linear routine.

In contrast, Table 9 and Figure 9 show that the control group's interaction was dominated by low-order cycling between Information Borrowing (IB) and Simple Agreement (SA). The strongest transitions were IB → SA ($Z = 14.0$) and SA → IB ($Z = 4.4$), indicating a self-reinforcing "transfer–endorse" loop in which AI- or peer-provided content was repeatedly carried forward and accepted with minimal scrutiny. Although the control group occasionally shifted from low-order behaviors to evidence-related activity (IB → ER, $Z = 6.2$; SA → ER, $Z = 5.2$), these transitions rarely developed into a stable higher-order sequence. Instead, evidence retrieval frequently regressed to low-order discourse (ER → IB, $Z = .3$; ER → SA, $Z = 2.1$), thereby disrupting progression toward sustained reasoning and regulation. Moreover, although ER → LA

Table 9. Residual transformation results of the control group.

Behavioral coding	IB	SA	QB	QB.AI	ER	ER.AI	LA	LA.AI	EJ	RA	CC
IB	1.5	14.0*	1.6	1.4	6.2*	1.5	1.7	1.1	1.2	0.9	0.7
SA	4.4*	1.4	1.8	1.5	5.2*	1.6	1.7	1.2	1.1	0.8	0.6
QB	1.4	1.3	−1.2	1.6	2.6*	1.4	1.5	1.2	1.1	0.9	0.7
QB.AI	2.1*	1.7	1.3	−1.1	1.2	1.1	1	0.8	0.7	0.6	0.5
ER	2.3*	2.1*	1.4	1.3	−1.1	1.2	2.4*	1.1	1	0.8	0.6
ER.AI	1.6	1.5	1.3	1.2	1.1	−1	1.3	1.2	1.1	0.9	0.7
LA	1.7	1.6	1.4	1.3	1.2	1.1	−0.9	1.1	1.3	1	0.8
LA.AI	1.5	1.4	1.2	1.1	1	0.9	1.1	−0.8	1	0.8	0.6
EJ	1.3	1.2	1.1	1	0.9	0.8	1	0.9	−0.7	0.7	0.5
RA	1.2	1.1	1	0.9	0.8	0.7	0.9	0.8	0.7	−0.6	0.4
CC	1.1	1	0.9	0.8	0.7	0.6	0.8	0.7	0.6	0.5	−0.5

*$p < 0.05$.

reached statistical significance ($Z = 2.4$), the subsequent pathway did not extend into evaluative judgment, reflective regulation, or consensus construction. Taken together, these patterns suggest that higher-order behaviors in the control group were episodic and weakly integrated, limiting their capacity to support systematic critical inquiry.Collectively, the LSA findings provide process-level evidence that metacognitive scaffolding in GenAI-mediated collaboration does not merely induce isolated critical behaviors; rather, it systematically reconfigures them into a coherent temporal mechanism. While the experimental group developed a robust, GenAI-specific higher-order sequence – originating from the auditing of AI outputs (QB.AI → ER.AI → LA.AI) and extending reliably toward evaluation, regulation, and consensus (EJ → RA → CC) – the control group remained entrenched in a low-order IB–SA loop, characterized by transient and easily disrupted excursions into deeper cognition. This sequential divergence supports the study's core premise that scaffolding functions as an epistemic regulator by reducing AI-related cognitive offloading and epistemic overreliance, thereby sustaining critical-thinking trajectories within GenAI-supported collaborative environments.

6. Discussion

This study elucidates the mechanisms through which metacognitive scaffolding cultivates critical thinking within Generative AI (GenAI)-supported collaborative learning environments. Our findings demonstrate that the experimental group significantly surpassed the control group across three key dimensions: critical thinking dispositions, high-order cognitive engagement, and the stability of sequential behavioral patterns. The following sections provide an in-depth discussion regarding the unique functional mechanisms by which metacognitive scaffolding catalyzes high-order cognition specifically within the GenAI-mediated pedagogical landscape.

6.1. Metacognitive scaffolding and the evolution of critical thinking in GenAI-supported environments

The findings of this study demonstrate that metacognitive scaffolds specifically tailored for GenAI environments significantly enhanced primary students' critical thinking performance, particularly within the dimension of "Questioning and Verification," which exhibited the most substantial effect size. This outcome transcends a mere quantitative increase in skill scores; rather, it elucidates the pivotal role of scaffolding in restructuring cognitive pathways during human-AI interaction.

From a theoretical perspective, the linguistic fluency of GenAI may encourage learners to adopt a more heuristic mode of processing, which can in turn increase the likelihood of cognitive offloading (Qi & Derakhshan, 2025; Vo et al., 2022). The "Plan-Monitor-Reflect" scaffolding designed in this study functions as a mechanism for introducing epistemic friction, effectively interrupting the automated trust typically afforded to AI-generated outputs. By forcing a shift from passive reception to active algorithmic auditing, the scaffolding compels learners to re-evaluate AI as a fallible interlocutor rather than an authoritative source of truth (Vijayakumar Bharathi & Pande, 2025). For sixth-grade learners, these findings suggest that adversarial prompting and AI-auditing did not impose an undue cognitive burden. Rather, the intervention was associated with higher-order engagement and more sustained questioning and verification, indicating that critical challenge can be introduced in ways that remain cognitively manageable for younger students. This finding further clarifies how metacognitive regulation operates in AI-augmented collaborative learning: the object of regulation is no longer limited to peer contributions, but increasingly includes epistemic vigilance toward AI-generated outputs.

Furthermore, these results challenge the prevailing view of AI as a "transparent" or "neutral" cognitive tool (Xu et al., 2025). The significant leap in the experimental group's ability to question and verify evidence underscores that critical thinking does not spontaneously emerge in the presence of advanced technology. Instead, it requires functional specificity in scaffolding – meaning that without interventions designed to counteract the "pseudo-authority" of AI, the potential for deep cognitive engagement remains unfulfilled. This aligns with and expands upon recent discourse regarding the necessity of human-centric agency in the age of generative intelligence.

6.2. Metacognitive scaffolding: reshaping the depth of cognitive engagement in GenAI-supported collaboration

The SOLO results suggest that metacognitive scaffolding deepened students' cognitive engagement by reshaping how they processed information during interaction with GenAI and peers. The experimental group showed a markedly higher proportion of relational and extended abstract discourse than the control group (49.7% vs. 27.3%), and the difference was especially pronounced at the extended abstract level (17.5% vs. 6.2%), with a significant overall between-group difference, $\chi^2(4, N = 1223) = 118.63$, $p < 0.001$, Cramer's $V = 0.31$. Rather than remaining at the level of fragmented response or surface acceptance, students under scaffolded conditions were more likely to critically examine AI-generated content, connect it with peer input and task demands, and develop more coherent and transferable reasoning. This suggests that metacognitive scaffolding promoted a progression from isolated understanding to relational meaning-making and reflective abstraction.

From a theoretical perspective, the linguistic fluency and perceived competence of GenAI systems may encourage more heuristic processing, prompting learners to outsource cognitive labor prematurely (Xie et al., 2025; Zheng et al., 2019). Our findings further suggest that, in human–AI collaborative ecosystems, cognitive depth is shaped not only by interpersonal knowledge negotiation, but also by how learners critically examine AI-generated content, connect it with peer input and task demands, and transform it into more coherent and transferable reasoning. Rather, it hinges critically upon the learner's "epistemic vigilance" toward AI-generated outputs (Rossouw & Steenkamp, 2025). By strategically embedding "epistemic friction" through structured planning, monitoring, and evaluation, these scaffolds deconstruct the "pseudo-authority" of GenAI, effectively transitioning the learner from a passive "information recipient" to an active "algorithmic auditor."

This conclusion challenges the techno-deterministic assumption that AI inevitably fosters "intellectual atrophy" or cognitive indolence(Feng, 2025; Yan et al., 2024). Diverging from pedagogical approaches that prioritize cognitive load reduction via simplified AI feedback (Yang et al., 2024), this study posits that "desirable difficulties" – engineered through scaffolded inquiry and verification – are indispensable for cultivating higher-order thinking. This conceptual transition from AI-related cognitive offloading and epistemic overreliance to cognitive reciprocity necessitates a reorientation of collaborative learning theories toward the systematic safeguarding of human agency.

6.3. Reconfiguring the process architecture: metacognitive scaffolding as an orchestrator of dynamic critical thinking systems in genai-supported collaboration

The ENA and LSA results converged to indicate that metacognitive scaffolding primarily reshaped the process architecture of critical thinking in GenAI-mediated collaboration, namely, the recurrent ways in which cognitive actions were organized, connected, and sustained over time. Its role therefore extended beyond improving the overall quality of students' thinking to restructuring how critical thinking was enacted as a coordinated epistemic process. The experimental group formed a denser and more coherent network centered on the auditing sequence of QB.AI → ER.AI → LA.AI, whereas the control group remained largely constrained within a comparatively fragile IB–SA loop. These differences point to two qualitatively distinct modes of collaborative cognition. Given that both groups used the same GenAI platform and received identical training, the divergence in discourse patterns appears more plausibly related to the scaffolding intervention than to a general novelty-driven elevation in engagement.

We argue that the scaffolding succeeds by instantiating a reusable critical-inspection routine (Borge et al., 2024). In the unscaffolded condition, interaction with GenAI lacks an explicit epistemic protocol, allowing a low-cost information-adoption and superficial-agreement script to dominate and to stabilize as a closed loop. These findings suggest that epistemic friction alone may be insufficient to sustain deeper reasoning. Its productive role appears to depend on whether scaffolding helps learners translate epistemic tension into questioning, verification, and reconstruction. By contrast, the present three-phase design (planning, monitoring, reflection) formalizes an alternative high-order routine (Perifanou & Economides, 2025; X. Zhou et al., 2024). It positions AI outputs as epistemic objects that warrant audit and specifies a canonical sequence of auditing operations: targeted questioning (QB.AI), external verification (ER.AI), and logical reconstruction (LA.AI). The robust sequential effects observed in LSA offer process-level traces that this routine was repeat-

edly activated and enacted within ongoing dialogue. In this sense, the scaffolding can be interpreted as an orchestration mechanism that helps translate abstract critical-thinking principles into executable, observable steps within human–AI interaction (Z. Zhou et al., 2025).

Taken together, these results suggest several implications for research on collaborative learning and critical thinking. First, they provide preliminary evidence for extending accounts of socially shared regulation to triadic human–machine–human collaboration, indicating that effective co-regulation may need to incorporate explicit appraisal of non-human epistemic contributions. Second, they advance the process-oriented agenda in the learning sciences by demonstrating how AI-related cognitive offloading/epistemic overreliance and cognitive reciprocity can be operationalized as interaction architectures, thereby yielding candidate process indicators for diagnosing and strengthening collaborative quality. Third, they refine debates on efficiency in human–AI collaboration by showing that perceived efficiency depends not only on fluency or rapid information exchange, but also on the governance of epistemic processes – specifically, whether groups sustain auditing and justification rather than defaulting to rapid agreement.

7. Conclusion

This study empirically explored the efficacy of metacognitive scaffolding in mitigating cognitive offloading, preliminarily revealing the potential for reshaping learners' engagement patterns within GenAI-supported collaborative environments. Our findings suggest that structured pedagogical interventions may effectively counterbalance the risks of superficial processing, guiding students from passive information adoption toward more deliberate cognitive involvement. Beyond validating a specific instructional tool, this research attempts to propose a viable pathway for safeguarding "cognitive agency" in the intelligent era. While this transition from instrumental dependence to cognitive reciprocity remains in its exploratory stages, it provides a foundational perspective for understanding how human agency can be institutionalized within human-AI symbiotic learning ecologies.

The findings offer several modest yet significant implications for the evolving discourse in educational technology. Theoretically, this study advances understanding of metacognitive regulation in GenAI-supported collaboration by showing that, in triadic Human–AI–Human interaction, the critical auditing of AI-generated outputs becomes a core component of collaborative regulation. In terms of pedagogical practice, the study invites designers to re-examine the balance between "technological fluency" and "deep learning," advocating for the strategic introduction of "epistemic friction" to stimulate critical vigilance. From a policy perspective, these insights offer a gentle call for educational stakeholders to broaden the definition of digital literacy – positioning algorithmic auditing as a critical component of digital equity. We hope these preliminary observations contribute meaningfully to the broader dialogue on responsible AI integration in education.

Despite the progress made, several boundary conditions warrant cautious interpretation. First, due to the quasi-experimental nature and the specific context of primary education, the "epistemic transferability" of these effects across different developmental stages or cultural backgrounds requires further validation through more diverse sampling. Second, as an initial exploration of GenAI collaboration, this study did not fully decouple the complex interaction effects between varying algorithmic logics (e.g. deterministic vs. stochastic feedback) and metacognitive regulation. Third, a further limitation concerns the possible novelty effect associated with learning in a GenAI-supported environment. Although both groups used the same platform and received the same training, the eight-week intervention may not have been long enough for the initial engagement and motivational boost associated with the new technological context to fully subside. Furthermore, the long-term "internalization effect" – whether students maintain heightened cognitive vigilance after the scaffolds are faded – remains a persistent question for future inquiry. Subsequent research should leverage longitudinal designs and multimodal learning analytics to refine our understanding of the evolving psychological mechanisms within human-AI co-evolution.

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Author contributions

Rongqing Zhuo: conceptualization, methodology, software, data curation, formal analysis, writing – original draft, writing – review and editing. Zhongxian Bai: conceptualization, methodology.

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No potential conflict of interest was reported by the author(s).

Ethics approval statement

This study was conducted under ethical guidelines. Each participant signed an informed consent form, and to ensure confidentiality, students' personal identifiers were removed prior to data processing.

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Data availability statement

The data are available upon request from the corresponding authors.

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